DitHub: A Modular Framework for Incremental Open-Vocabulary Object Detection

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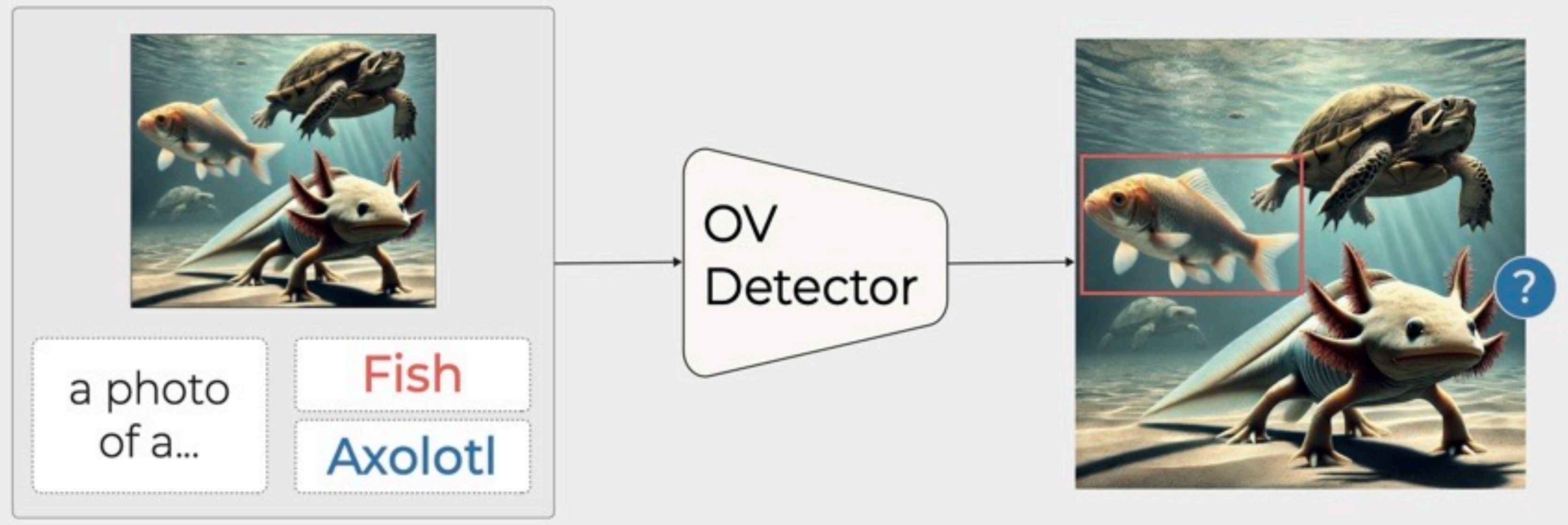
Neural Information Processing Systems

Intro: Open-Vocabulary Object Detection

Open-Vocabulary Object Detection allows models to detect objects from text prompts beyond fixed label sets.

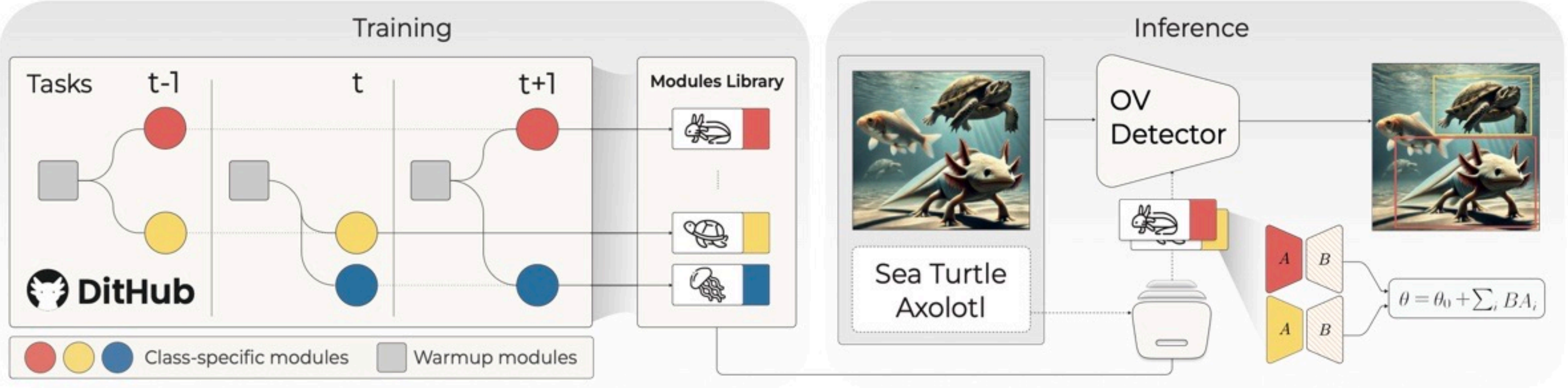
However, real applications still require **targeted adaptation** to specialized domains, rare classes, and distribution shifts — while preserving zero-shot recognition.



TL;DR

Modularity: A complex software system is not built from a single block of code, but from modules developed at different times and by different contributors.

DitHub: An open-vocabulary detection framework built on class-specific expert LoRA modules, governed by a structured lifecycle of branching, fetching, and merging.

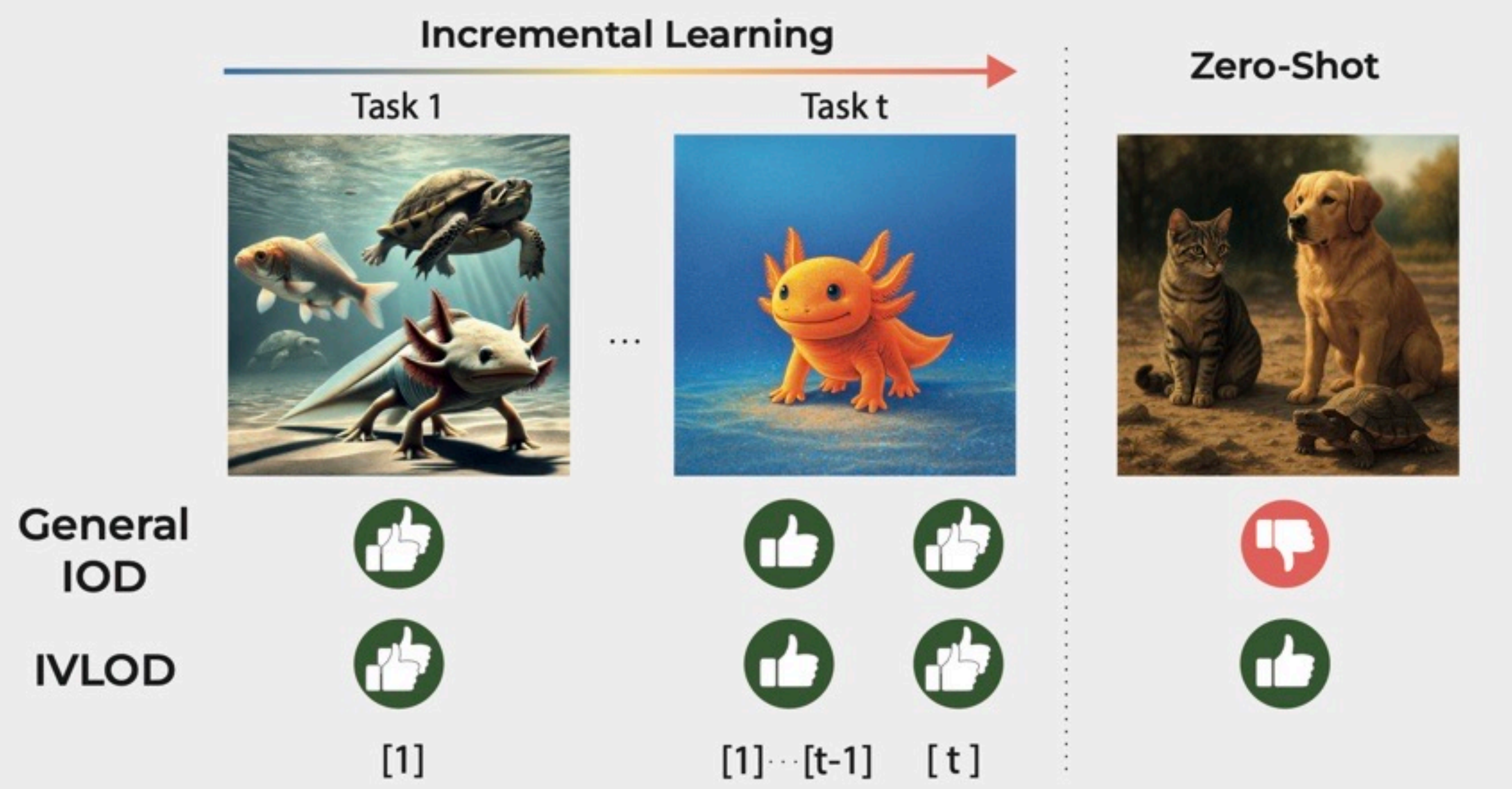


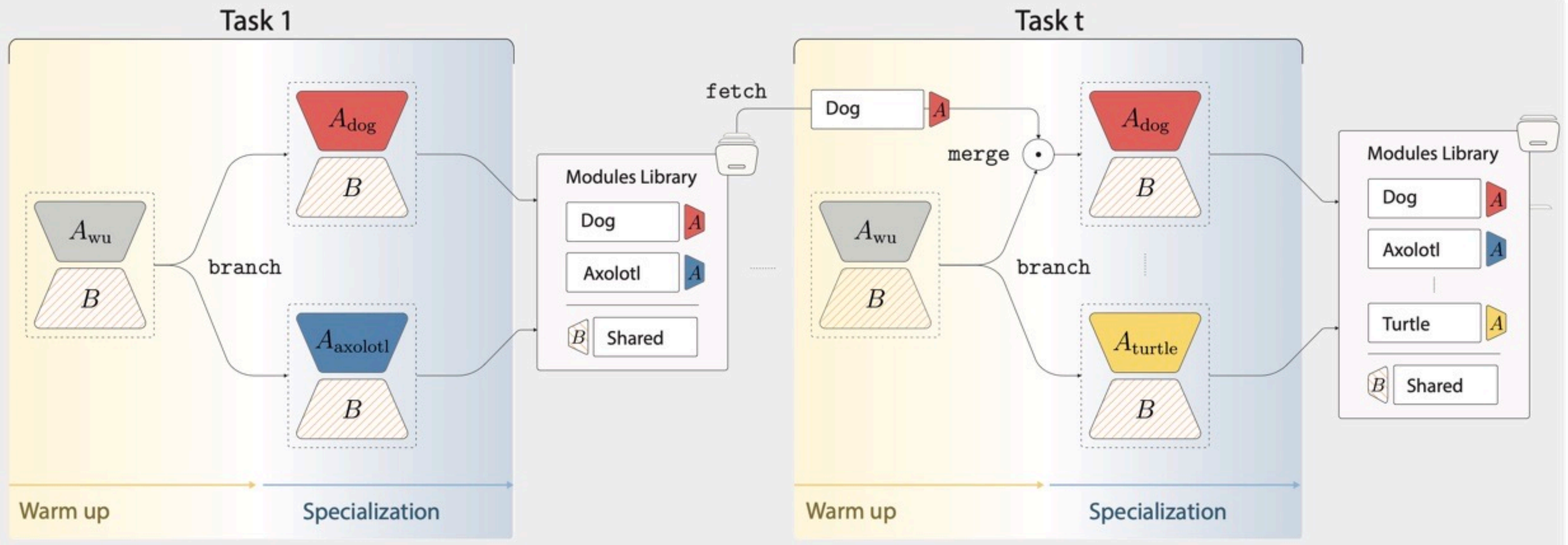
At **inference**, DitHub retrieves the experts linked to the text query and composes them into a unified LoRA module for precise detection.

Problem statement: IVLOD

Incremental Vision-Language Object Detection (IVLOD) extends open-vocabulary detection by enabling models to learn new classes over multiple tasks while preserving zero-shot recognition.

It addresses real scenarios where object categories appear gradually and may shift across domains.





Warm up

Learn a single task-level module to provide a **shared initialization** for all class experts.

Specialization

Create one module per class and refine each independently to learn **class-specific** features.

DitHub Primitives

`branch`: Create one expert module per class for independent specialization.

`fetch`: Retrieve the previous expert of a class when it reappears in a new task.

`merge`: Weighted interpolation of old and new modules. $B_t = (1-\lambda)B_{t-1} + \lambda_B B^{opt}$

Performance on ODinW-13 and ODinW-O

This evaluation demonstrates DitHub's ability to effectively retain knowledge in incremental learning scenarios while excelling in zero-shot detection, setting a new **state-of-the-art**.

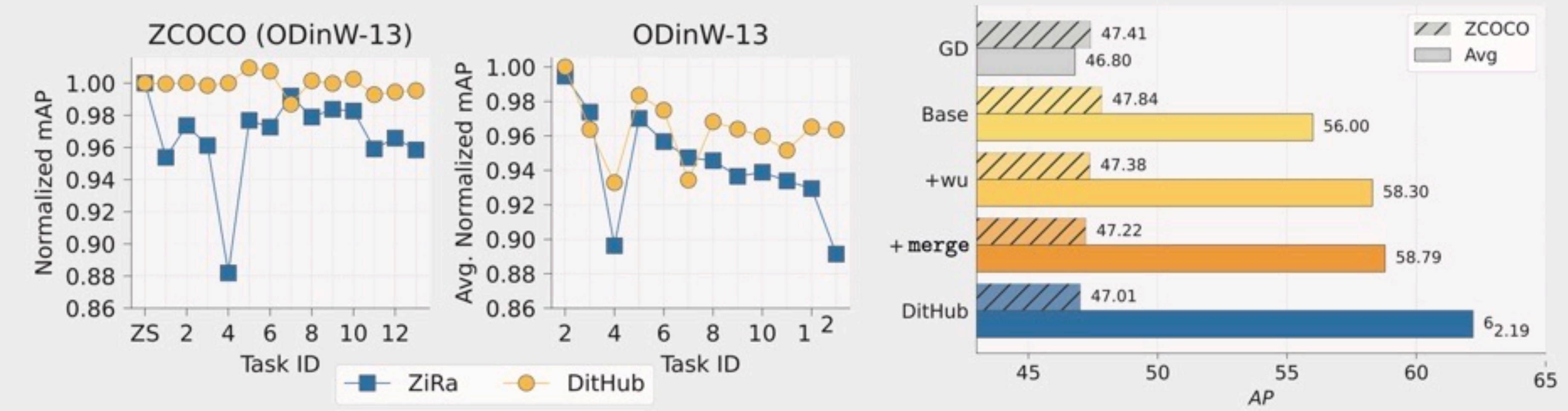
Method	ZSCOCO	Avg	Ae	Aq	Co	Eg	Mu	Pa	Pv	...
G-DINO	47.41	46.80	19.11	20.82	64.75	59.98	25.34	56.27	54.80	...
ZiRA	46.26	57.98	31.76	47.35	71.77	64.74	46.53	62.66	66.39	...
DitHub	47.01	62.19	34.12	50.65	70.46	68.56	49.28	65.57	69.58	...

We introduce **ODinW-O** (Overlapped), a new benchmark based on ODinW-35, focusing on classes that **reappear across multiple tasks**.

Method	ZSCOCO	Avg	Ae	Hw	Pv	Sd	Th	Ve
G-DINO	47.41	53.15	45.12	67.54	58.11	25.84	70.40	51.87
ZiRA	44.43	57.63	39.92	68.00	64.90	46.26	77.26	49.47
DitHub	46.51	62.38	53.35	71.07	71.01	41.75	80.21	56.90

DitHub preserves zero-shot ability while improving incremental adaptation

Ablation study



DitHub class-unlearning "horse"